

Abbas Quettawala

PROFESSIONAL SUMMARY

Final-year Mechatronics Engineering (Honours) student at Monash University with hands-on experience building and integrating robotic hardware, from CAD and 3D-printed parts to motors, electronics and embedded control. Combines mechanical assembly, soldering and practical testing with robotics software and reinforcement learning, with a strong interest in human-centric robots.

EXPERIENCE

Undergraduate Studentship | CSIRO

Jun 2025 – Present

- Collaboratively developed ManuMentor, a manufacturing retrieval-augmented generation (RAG) assistant combining technical manuals, structured test data and subsystem knowledge graphs for source-backed answers and guided troubleshooting.
- Worked collaboratively on a Vue 3 and Python/FastAPI platform integrating OCR-based document ingestion, vector search and Memgraph knowledge graphs, with interactive checks to guide engineering fault investigation.
- Co-authored a research paper and assisted with the project's ethics trials, communicating technical work through research documentation.

Algorithmics Developer | Orica (Monash IIP placement)

Nov 2025 – Apr 2026

- Manually annotated raw drone footage with bounding boxes, augmented labelled data using Python, Albumentations, OpenCV and NumPy, and trained and compared YOLO object-detection models; integrated augmentation and mosaic stitching into a PyQt6 prototype demonstrated to Orica stakeholders.
- Assisted with drone electronics assembly and supported flight planning using ArduPilot, PX4 and ROS 2.

Mechatronics Engineer | Monash Automation

Apr 2026 – Present

- Own reinforcement-learning and simulation development within a multidisciplinary student robotics team; share physical prototyping and hardware integration with teammates.
- Support prototype design, soldering, mechanical assembly and electronics integration; document testing and technical findings and contribute to demonstrations.
- Support lab maintenance, including P1S printers with AMS and Prusa XL 3D printers.
- Use KUKA robot arms, camera systems and LiDAR across team projects.

Project Team Member / RL Project Lead | Monash Deep Neuron

Mar 2023 – Apr 2026

- Led the project developing reinforcement-learning agents for Tetris, Sonic and Rocket League.
- Helped develop Bookworm GPT to turn PDFs, PowerPoint slides and screenshots into student notes.
- Built a skin-lesion identification pipeline from scratch, created its image dataset, and trained and validated computer-vision models.
- Designed and built a GAN from scratch, created character-image datasets, and trained and validated models to generate Pokémon and Animal Crossing characters.

PROJECTS

Waddle (Bipedal Robot) | Monash Automation

- Owned the reinforcement learning and simulation pipeline for a 10-joint biped; trained phase-conditioned PPO across 512 Isaac Lab environments to produce alternating forward steps in simulation, diagnosing actuation/frame faults and documenting standing and gait validation.
- Assisted with CAD design, mechanical assembly and electronics integration, including CubeMars motor installation and CAN bus communication.

Ten 6-DOF Robotic Arms | Monash Automation

- Collaboratively designed and built ten 6-DOF arms adapted from FABRI CREATOR, with custom 3D-printed bases, ESP32 control and LiPo power; developed a custom control protocol for touchscreen-triggered emotes and demonstrated the arms to senior engineering executives at the Engineering Dean's Lunch showcase.

PROJECTS (CONTINUED)

Hollow Knight Reinforcement Learning Agent | Personal project

- Implemented PPO from scratch in PyTorch and built a custom C# mod for a Hollow Knight reinforcement learning (RL) agent; trained using four simultaneous game instances, with over 200 hours of GPU-accelerated training.

ESP Mini Drones | Personal project — in development

- Designing an ESP32-based mini quadcopter, evaluating hardware and firmware options for onboard sensing and flight control.

AI Desk Buddy | Personal project — in development

- Developing a Raspberry Pi 4 desk companion with a Pi Camera for face tracking and PC-assisted AI conversation.

Homelab & Server Automation | Personal project — in development

- Building a Proxmox homelab on a repurposed PC; configuring Hermes and cron jobs for news updates, maintaining computers and servers, and developing a Kubernetes deployment.

MuseMender (Sheet Music to MuseScore) | Personal project

- Built a Python/PySide6 tool converting printed piano scores, PDFs and sheet-music videos into editable MuseScore files.

Embedded Systems Projects | Monash University

- Designed a 16-bit FPGA processor in Verilog.
- Built an ultrasonic distance sensor using PSoC and C.
- Built a camera-based water-level system using Arduino and Python.
- Built a custom C program to scan and decode item barcodes across varying angles and lighting conditions.

TECHNICAL SKILLS

Robotics & AI: ROS 2, Isaac Sim/Lab, URDF, reinforcement learning (PPO, DQN), PyTorch/TensorFlow

Embedded & Design: ESP32, Arduino, PSoC, FPGA/Verilog, SolidWorks/Onshape, 3D printing, servos, LiPo power

Software & Testing: Python, C, C#, JavaScript, OpenCV, NumPy, FastAPI, PyQt6, Git, Linux; unit tests, simulation test protocols

ADDITIONAL EXPERIENCE

VCE Mathematics Tutor | A4 Maths Learning Centre

Jul 2023 – Present

- Tutor Years 4–12, including VCE Methods, Specialist Mathematics and Physics; tailor extension tasks, exam practice and CAS strategies.

Fresh Convenience Team Member | Woolworths

Jan 2023 – Feb 2026

- Managed chilled stock and deli service, applying FIFO rotation, hygiene and freshness checks.

Personal Maths Tutor | Self-employed

Jan 2022 – Dec 2025

- Tutored Years 9–12 through tailored maths practice, problem walkthroughs and SAC/exam preparation.

Coding Tutor | Digimaker

Feb 2025 – Jul 2025

- Taught Python, JavaScript and Scratch to Years 3–6 through project-based programming lessons.

Customer Liaison | Target

Dec 2019 – Jan 2022

- Handled sales, refunds and pricing queries; closed tills, replenished stock and assisted customers.

EDUCATION

Bachelor of Mechatronics Engineering (Honours) | Monash University

2022 – expected completion mid-2027

Forest Hill College

2016 – 2021

VCE (2021); ATAR 92.15; School Dux; ranked first in Specialist Mathematics and Accounting.